

# First Demonstration of a Fully-Vertical GaN Power finFET with Direct Optical Triggering

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**Abstract**—Optically controlled power devices hold immense promise for enabling next-generation high-voltage systems with enhanced performance and simplified architectures. In this work, we present a fully vertical, optically triggered GaN finFET directly driven by a UV LED. The device achieves a remarkable optical gain of  $R > 10^4$  A/W, a robust current on-off ratio of  $I_{on}/I_{off} > 10^7$ , and a low specific on-resistance of  $R_{on,sp} = 60$  mΩ cm<sup>2</sup> under an optical power density as low as  $P_{LED} = 17$  mW/cm<sup>2</sup>. Furthermore, it demonstrates an off-state breakdown voltage  $BV > 600$  V. A single, large area device with  $R_{on} = 2.6$  Ω was also successfully demonstrated. These results highlight the significant potential of this device to mitigate electromagnetic interference (EMI) and simplify gate driver design in high-voltage electronics, paving the way for cost-effective, high-efficiency optically controlled power systems with innovative topologies.

**Index Terms**—Gallium Nitride, Power Transistors, finFETs, EMI, Phototransistors, Optical Control, Vertical Device

## I. INTRODUCTION

THE rising demand for cloud computing and electric vehicles has highlighted the importance of efficient power delivery and transformation through the grid. Although the development of wide-bandgap semiconductors has enabled single devices with high breakdown voltage, for medium-high-voltage applications (5-50 kV) such as distribution grids, solid-state transformers or electric train propulsion, the use of multilevel converters is often necessary [1], [2]. Using electrically triggered power modules in these circuit topologies can pose significant challenges in the design of gate driving circuitry and concerns in electromagnetic interference (EMI) [3], [4]. By using optically controlled power devices, gate driving circuitry can be simplified with improved EMI susceptibility (Fig. 1) [5], [6]. GaN-based optically-triggered power devices have been demonstrated with photoconductive switches (PCSS). However, they are limited by their low optical gain, requiring complex high-power pulsed UV lasers to achieve low on-resistance. Thus, the use of PCSS has been limited to pulsed-power applications [7]. There are other attempts at using optical transistors [8] or photodiode [9] to drive high-voltage

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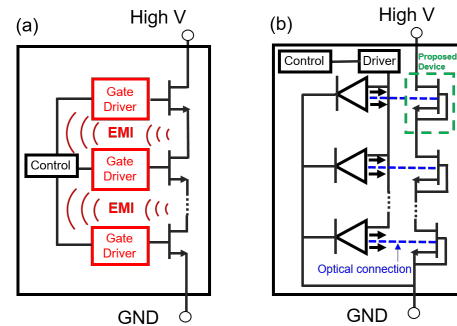


Fig. 1. Comparison of (a) Conventional electrically triggered modules with complex high-side drivers and significant EMI and (b) Proposed optically triggered design with simplified stacking and improved EMI.

power devices. Nevertheless, these approaches cannot address challenges of source-referencing in a cascoded structure and would still need a gate driver for the main switches.

In this work, we demonstrated for the first time, fully vertical, optically controlled GaN finFETs on bulk GaN substrates with 6 orders of magnitude improvement in the experimental on-off current ratio ( $10^7$ ) compared to the previous state of the art [10]. The devices exhibit high optical gain ( $10^4$  A/W), low  $R_{on,sp}$  (60 mΩ cm<sup>2</sup>) and high breakdown voltage (623 V). Initial measurements of the switching response for these devices were also performed, and a fin width-dependent asymmetric transient behavior was identified with sub-ms turn-on delay. These measurements highlight that the proposed device exhibits significant potential for applications such as DC circuit breakers, where the requirement for fast turn-off is less stringent.

## II. DEVICE STRUCTURE AND FABRICATION

Fully vertical, optically-controlled GaN finFETs with various fin widths were fabricated on conductive bulk GaN substrates from Ammono. The epitaxial structure was grown using metal-organic chemical vapor deposition (MOCVD). The fabrication process, illustrated in Fig. 2, begins with the formation of fin channels, followed by the deposition of field oxide (SiO<sub>2</sub>), source metal stack (Ti/Al), and gate oxide (Al<sub>2</sub>O<sub>3</sub>). The use of plasma-enhanced atomic layer deposition (PE-ALD) was employed to minimize positive interface

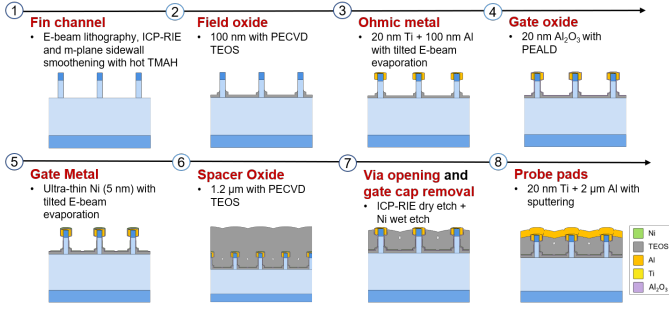


Fig. 2. Process flow of fully-vertical, optically controlled GaN finFET.

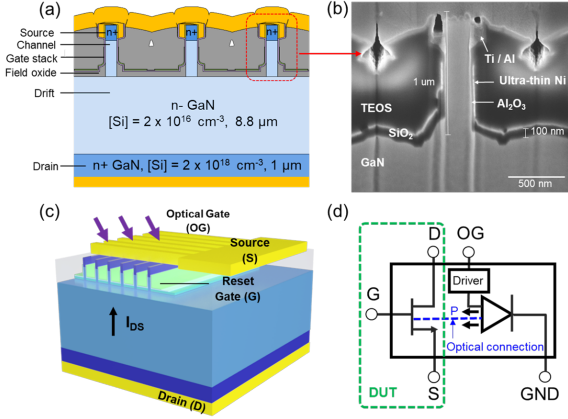


Fig. 3. (a) Cross-sectional schematic and (b) Focused Ion Beam (FIB)-Scanning Electron Microscopy (SEM) cross-section image of an exemplary fabricated device. (c) 3D diagram of the proposed device and (d) Circuit schematic illustrating the terminals of the proposed optical module. OG: Optical Gate Driver Signal. P: Optical Power.

charges, ensuring normally-off device operation [11]. To fully deplete the fin channels in the dark, an ultra-thin (5 nm) nickel electrode (G) connected to the source (S) during optical measurements was utilized to establish the surface potential on the fin sidewalls and deplete the channel. A UV-transparent oxide served as the fin spacer, and the process concluded with the deposition of drain metal and probe pads. Fig. 3(a) and (b) show the cross-sectional schematic and SEM image of an exemplar device, respectively, while Fig. 3(c) provides a 3D diagram illustrating interdigitated probe pads to provide optical windows. The optical gate power (P) is controlled by the optical source driver (OG). Fig. 3(d) illustrates the circuit schematic for the proposed integrated optical module.

### III. RESULTS AND DISCUSSION

#### A. Operation Principle

The proposed device comprises two electrical terminals: source (S) and drain (D). During optical switching, the reset gate (G) is connected to the source terminal. In the absence of illumination, the high work function of the gate metal depletes the fin channel, leading to a low dark current, similar to that of a normally-off finFET [12]. Upon illumination, the fin channel absorbs incident photons, generating electron-hole pairs. These charge carriers are separated by the internal electric field,

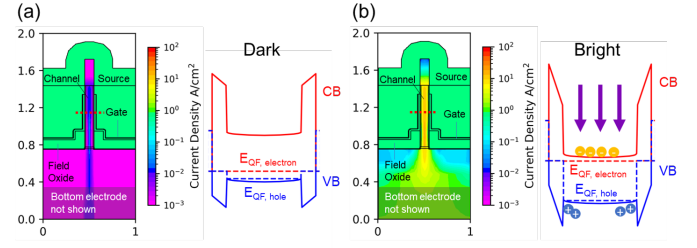


Fig. 4. Simulated current density and band diagram of the proposed device under dark and illuminated states.

leading to a high optical gain. Turning off the device requires the removal of holes from the channel, typically through recombination with electrons or drift driven by the electric field. This process introduces a trade-off between optical gain and turn-off time, which will be analyzed in detail in a subsequent section. Additional insights into the operation principles are provided in our previous work [10]. Fig. 4 illustrates the simulated current density and band diagrams of the device under dark and illuminated conditions. These simulations were performed using Silvaco Victory, with Auger, Shockley-Read-Hall (SRH), and optical recombination models activated.

#### B. Static Performance

The devices were measured using an Agilent B1505 semiconductor parameter analyzer. The optical control was provided by a 365 nm collimated fiber-coupled UV LED. All measurements were normalized to the active area of the device ( $250 \mu\text{m} \times 250 \mu\text{m}$ ). Fig. 5(a) presents the transfer characteristics of a device with a fin width of 150 nm under both illuminated and dark conditions. At  $V_{GS} = 0 \text{ V}$ , the on-off current ratio exceeds  $10^7$ , 6 orders of magnitude higher than our previous work [10]. This huge improvement is attributed to PE-ALD deposited gate oxide that ensures a positive threshold voltage without illumination [11]. Fig. 5(b) shows the output characteristics of the same device. A record-low  $R_{on,sp}$  of  $60 \text{ m}\Omega \text{ cm}^2$  is achieved for the 150-nm-fin device under an optical irradiance of just  $17 \text{ mW/cm}^2$ . This corresponds to a high optical gain,  $R > 10^4 \text{ A/W}$ , as illustrated in Fig. 5(c), where  $R$  is defined as the ratio between the drain current and the input optical power. The  $\alpha$  coefficient, where  $R \propto P^{\alpha-1}$ , is extracted to be approximately 0.7, suggesting that  $R_{on,sp}$  can be further improved with increased optical power. Fig. 5(d) summarizes the effect of fin width on device performance. The threshold voltage decreases with fin width, resembling the behavior observed in JFETs. Additionally, the maximum on-current at  $V_{GS} = 0 \text{ V}$  under illumination decreases with fin width due to a reduced effective channel conduction volume caused by increased surface depletion from the fin sidewalls. An optimum on-off current ratio is observed for a fin width of 150 nm.

A  $1 \text{ mm} \times 1 \text{ mm}$  device with 200-nm fins was also successfully demonstrated. As shown in Fig. 6, this device achieves over 1 A of pulsed drain current at  $V_{DS} = 5 \text{ V}$ ,

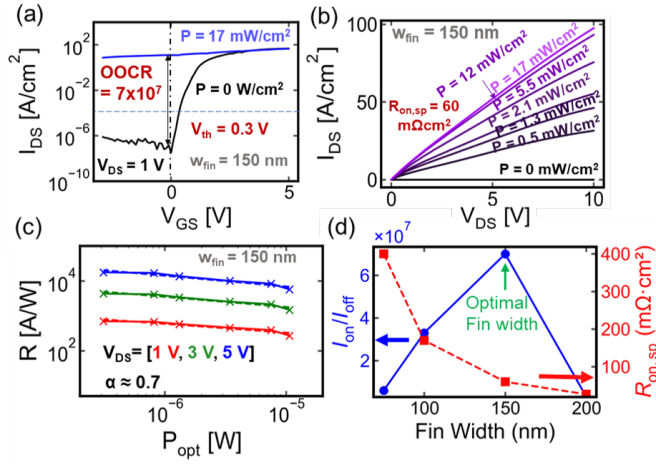


Fig. 5. DC response of the fabricated optical power switches. (a) Transfer characteristics of devices with 150 nm fin width under dark and illuminated conditions.  $V_{DS} = 1$  V. (b) Output characteristics with different light intensity in the optical mode ( $V_{GS} = 0$  V). (c) Responsivity as a function of optical power under different  $V_{DS}$  with  $\alpha$  coefficient extracted. (d) On-off current ratio and  $R_{on,sp}$  as a function of fin-width at  $P = 17$  mW/cm<sup>2</sup>.

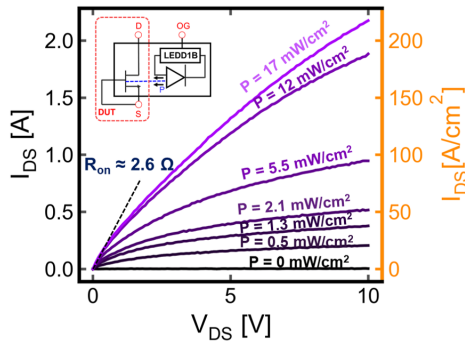


Fig. 6. Optical output characteristics of a large area device with an active area of 1 mm<sup>2</sup>. Fin width = 200 nm.

triggered by a UV LED when  $P = 17$  mW/cm<sup>2</sup>. The resulting  $R_{on,sp}$  of the 200-nm fin, large-area device closely matches that of the small-area device in Fig. 5(d), confirming the uniformity of the collimated light beam.

### C. Dynamic Performance

Transient measurements were conducted to evaluate the dynamic response of the optical switch. A constant  $V_{DS}$  of 5 V was applied to the fabricated devices, and the resulting current was monitored using a semiconductor parameter analyzer. The UV LED excitation signal was modulated by an LED driver, controlled by a waveform generator.

As illustrated in Fig. 7(c), the devices demonstrate fast turn-on behavior, which can be further accelerated by increasing the light intensity to enhance the gate photocurrent. The turn-off is relatively slow, primarily due to difficulties in removing photogenerated carriers. This effect is more pronounced in devices with wider fins, which exhibit higher optical responsivity facilitated by lateral field-assisted carrier separation. While the lateral field effectively lowers on-resistance during operation,

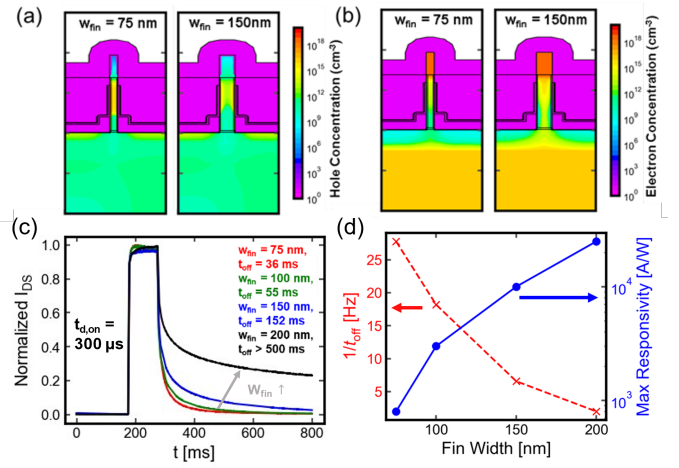


Fig. 7. Simulated (a) hole and (b) electron concentration in the fin channels under illumination (on state). (c) Normalized current transients under an optical pulse. The turn-off time ( $t_{off}$ ) is defined as the time for the current to fall to 10% of the peak value, and the turn-on delay ( $t_{d,on}$ ) is defined as the time for the current to rise to 10% of the peak value. Measurement conditions:  $V_{GS} = 0$  V,  $V_{DS} = 5$  V. (d)  $1/t_{off}$  and maximum responsivity as a function of fin-width.

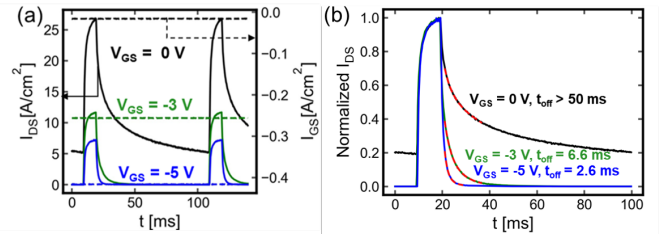


Fig. 8. (a) Nominal current density under a 10 Hz optical waveform. Solid:  $I_{DS}$ , dashed:  $I_{GS}$ . (b) Normalized current transient with extracted characteristic turn-off time ( $t_{off}$ ).  $t_{off}$  is significantly reduced by electron injection through the gate. Fin width = 100 nm.

it also reduces the probability of carrier recombination once the light source is switched off. In contrast, devices with narrower fins (75 nm) achieve significantly faster turn-off times, as the overlapping spatial distribution of electrons and holes enhances recombination efficiency, as shown in Fig. 7(a) and Fig. 7(b). The trade-off between frequency response and gain, depicted in Fig. 7(d), is a characteristic feature commonly observed in nanowire-based photodetectors [13].

To study the origin of the slow turn-off, gate-controlled transient measurements were performed by applying negative biases to the reset gate terminal (G), as shown in Fig. 8. This approach reduced the turn-off time significantly by injecting electrons through the gate to remove the photogenerated holes, albeit at the expense of reduced on-current. These results underscore the importance of efficient minority carrier removal in achieving fast photoresponse. To eliminate the need for an external gate bias, a potential solution involves incorporating p-type regions into the channel. This would form an intrinsic p-n diode, facilitating the removal of holes.

Despite their slow turn-off characteristics, the MOS-based



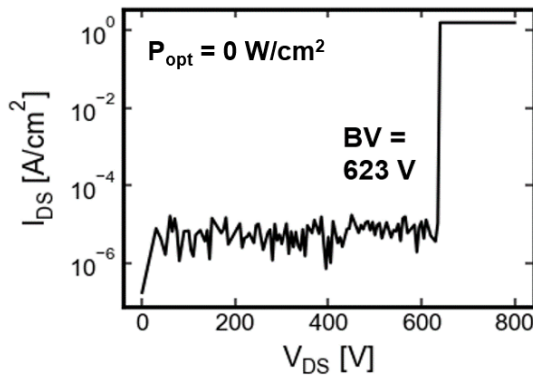


Fig. 9. Breakdown measurement of an exemplary device.

optically triggered GaN finFETs are well-suited for applications where rapid turn-off is not required. One promising application lies in thyristor-based DC circuit breakers [14], where these devices can compete effectively with light-triggered silicon thyristors. The combination of high critical field, robust high-temperature operation, and low on-resistance under illumination provides distinct advantages, positioning these devices as a compelling alternative in high-voltage power electronics.

#### D. Breakdown Characteristics & Benchmarking

The breakdown characteristics were measured in the off state without illumination. As shown in Fig. 9, the breakdown voltage (BV) exceeded 600 V. It can be further improved through proper edge termination or reduced drift doping. Lastly, the fabricated devices were benchmarked against other wide band-gap, optically controlled power devices (Fig. 10). Our devices demonstrated low on-resistance under LED illumination, offering a promising pathway towards low-cost, all-optical, wide-bandgap device modules and solid-state circuit breakers.

### IV. CONCLUSION

In this work, we demonstrated a fully-vertical, optically triggered GaN finFET with an on-off current ratio greater than  $10^7$  using a low-power UV LED. The fabricated devices exhibited high photoresponsivity ( $10^4$  A/W), low specific on-resistance  $R_{on,sp} = 60 \text{ m}\Omega\text{cm}^2$ , and fast turn-on. Additionally, a BV of 623 V was achieved. The proposed device was benchmarked against other optical power switches, highlighting its potential in future low-cost, all optical, high-voltage power modules with reduced EMI susceptibility and circuit complexity.

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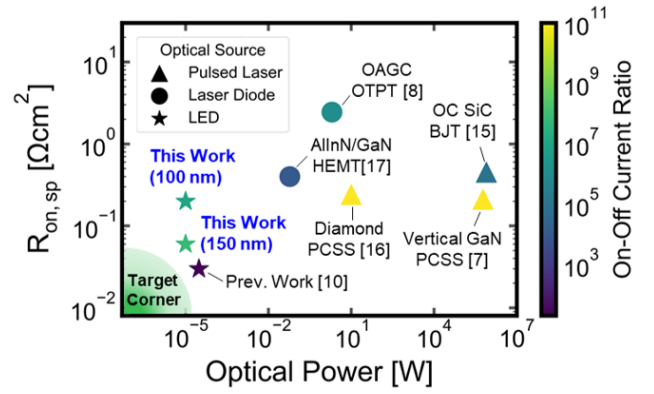


Fig. 10. Benchmarking plot against other demonstrations of wide band-gap, optically-controlled power switches. [7], [8], [10], [15]–[17]

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